When the Same Question Gets Different Answers:

Quantifying Non-Determinism in LLM-Assisted Evidence Synthesis for Environmental Health Research

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Abstract

Background. Large language models (LLMs) are increasingly adopted to assist systematic reviews and evidence synthesis, performing tasks such as abstract screening and data extraction. A critical but underexplored assumption is that these models produce consistent outputs when given identical inputs and configurations.

Objective. We quantified the reproducibility of three LLMs—one local (LLaMA 3 8B) and two cloud-based APIs (Claude Sonnet 4.5, Gemini 2.5 Pro)—across a complete evidence synthesis pipeline for $PM_{2.5}$ and respiratory health research.

Methods. We constructed a 500-abstract corpus from PubMed (100 include, 100 exclude, 300 borderline cases selected to stress-test decision consistency) and ran each model 10 times under identical configurations (temperature = 0, fixed seeds where supported) through two stages: abstract screening (500 abstracts per run) and structured data extraction (100 included articles per run). Reproducibility was measured using the Exact Match Rate (EMR)—the proportion of items receiving identical outputs across all 10 runs—with 95% bootstrap confidence intervals (10,000 resamples).

Results. The locally deployed LLaMA 3 achieved perfect reproducibility across all successfully processed items (EMR=1.000) in both screening and extraction. Cloud-based APIs exhibited substantial non-determinism: Claude Sonnet 4.5 achieved screening EMR=0.974 [0.960, 0.986] but extraction EMR=0.050 [0.010, 0.090]; Gemini 2.5 Pro achieved screening EMR=0.936 [0.914, 0.956] and extraction EMR=0.200 [0.120, 0.280]. Despite requesting deterministic behavior, 95% of Claude's and 80% of Gemini's

extraction outputs varied across runs. Semantic equivalence analysis—including BERTScore embedding similarity (F_1=0.997), Levenshtein fuzzy matching, and text normalization—confirmed that metadata text is near-identical across runs, yet the numeric extraction layer (effect sizes, confidence intervals) drives the low hash-based EMR. Meta-analytic propagation analysis revealed that 21% of articles yielded a different number of effect estimates across runs, causing the composition of a downstream meta-analysis to depend on which LLM run generated its inputs.
Conclusions. LLM non-determinism poses a concrete threat to the reproducibility of evidence synthesis, from individual study extraction through to pooled meta-analytic conclusions. Researchers using cloud-based LLMs should expect meaningful output variation even under "deterministic" settings and adopt provenance-tracking protocols to document and mitigate this instability.

Keywords: reproducibility, large language models, evidence synthesis, systematic review, non-determinism, $PM_{2.5}$, screening, data extraction

Contents

1 Introduction

1.1 The Promise of AI in Evidence Synthesis

Evidence synthesis—the systematic process of identifying, appraising, and combining research findings—forms the backbone of evidence-based medicine and public health policy. Systematic reviews and meta-analyses sit atop the hierarchy of evidence, informing clinical guidelines, regulatory decisions, and resource allocation worldwide [Higgins et al., 2019, Page et al., 2021]. Yet the process is notoriously labor-intensive: a typical systematic review requires 6–18 months and hundreds of person-hours, with title and abstract screening alone consuming an estimated 25% of total effort [Borah et al., 2017].

The emergence of large language models (LLMs) has created unprecedented opportunities to accelerate this process. Recent studies demonstrate that models such as GPT-4 [Achiam et al., 2023], Claude [Anthropic, 2024], and Gemini [Comanici et al., 2025] can screen abstracts with sensitivity approaching that of human reviewers [Oami et al., 2024, Khraisha et al., 2024], extract structured data from published studies with over 90% accuracy [Gartlehner et al., 2024, Jensen et al., 2025], and even assist in evidence quality assessment [Wang et al., 2025, Li et al., 2025b]. A growing body of work in the *Journal of the American Medical Informatics Association*, *Research Synthesis Methods*, and *npj Digital Medicine* has established LLM-assisted evidence synthesis as a legitimate and rapidly expanding methodological frontier [Li et al., 2025a, Scherbakov et al., 2025, Matsui et al., 2024].

1.2 The Hidden Problem: Non-Determinism

Beneath this promise lies an assumption so fundamental that it is rarely stated explicitly: *if you ask the same question twice, you should get the same answer*. In the context of LLM-assisted evidence synthesis, this means that running the same model with the same prompt, temperature, and random seed on the same abstract should produce identical screening decisions and extraction outputs every time.

This assumption does not hold in practice.

Recent work has demonstrated that commercial LLM APIs exhibit non-deterministic behavior even when configured for maximum reproducibility. Atil et al. [2024] showed accuracy variations of up to 15% across repeated identical runs, with best-to-worst performance gaps reaching 70%. Klishevich et al. [2025] confirmed that temperature $= 0$ does not guarantee deterministic outputs across any of the major models tested (GPT-4o, Claude 3.5, LLaMA 3.2). At a deeper level, Fu et al. [2026] demonstrated that even when surface-level text appears identical, the underlying token probability distributions exhibit substantial instability, suggesting that apparent determinism may mask hidden variation. A philosophical analysis by Karelin et al. [2024] traced this non-determinism to three fundamental sources: numerical precision in floating-point operations, architectural choices in distributed inference, and sampling procedures inherent to the generation process.

1.3 Why This Matters for Evidence Synthesis

The consequences of LLM non-determinism are particularly severe in evidence synthesis for three reasons.

First, decisions propagate. A screening decision that flips from "include" to "exclude" in a repeated run does not merely change one data point—it removes an entire study from downstream analysis. In meta-analysis, where pooled effect estimates are sensitive to study composition, a single inclusion change can shift a confidence interval across the null, transforming a statistically significant finding into a non-significant one [Rover, 2026].

Second, extraction compounds variation. Data extraction involves free-text parsing of complex scientific results—relative risks, confidence intervals, exposure definitions, population characteristics. Even small variations in how an LLM parses "RR = 1.05 (95% CI: 1.01–1.09)" can propagate through meta-analytic pooling. Gartlehner et al. [2024] reported 95–97% test-retest reliability in LLM extraction, but this still implies 3–5% of extracted values may change between runs—a rate that, across hundreds of data points, can materially alter pooled estimates.

Third, reproducibility is the foundation of scientific credibility. The broader reproducibility crisis in AI research [Hutson, 2018, Haibe-Kains et al., 2020, Kapoor & Narayanan, 2023] has already eroded trust in computational findings. When LLM-assisted reviews cannot be independently reproduced because the API returns different outputs, the scientific value of those reviews is undermined [Han, 2025, Desai et al., 2025, Semmelrock et al., 2025].

1.4 The Gap We Address

Despite the rapid adoption of LLMs in evidence synthesis, **no study has systematically measured how API-level non-determinism propagates through a complete synthesis pipeline**—from abstract screening through data extraction—in a real health domain.

This gap is not a minor omission; it represents a fundamental blind spot. Consider: Jensen et al. [2025] report 94.1% inter-session reproducibility for ChatGPT-4o extraction—an encouraging number. But this was measured with only 2 repetitions and evaluated per-field, not per-output. When we apply a stricter whole-output metric across 10 repetitions, we find extraction EMR as low as 5.0%—nearly twenty times worse than what per-field metrics suggest. Similarly, Gartlehner et al. [2024] report

95–97% test-retest reliability, but with only 2–3 repetitions and no examination of how variation compounds across pipeline stages. Studies measuring non-determinism [Atil et al., 2024, Klishevich et al., 2025] do so on general NLP benchmarks, where the structured, multi-field outputs characteristic of evidence synthesis are absent. And the most influential screening evaluations [Oami et al., 2024, Khraisha et al., 2024, Matsui et al., 2024] report results from a single run, implicitly assuming that a repeat would yield the same result.

We bridge this gap by:

1. Quantifying reproducibility across 10 repeated runs of 3 LLMs (1 local, 2 API) through a two-stage evidence synthesis pipeline;
2. Measuring variation at multiple granularities: whole-output, per-field, and per-abstract;
3. Comparing local versus cloud-based deployment as a determinism strategy;
4. Demonstrating that common "deterministic" API settings (temperature = 0, fixed seeds) are insufficient to ensure reproducibility.

2 Related Work

2.1 LLMs in Systematic Reviews

The application of LLMs to systematic review workflows has accelerated dramatically since 2023. Oami et al. [2024] evaluated GPT-4 Turbo for citation screening in *JAMA Network Open*, reporting sensitivity of 0.75 and specificity of 0.99, with prompt tuning improving sensitivity to 0.91. Khraisha et al. [2024] extended this to multilingual screening and extraction in *Research Synthesis Methods*, finding strong specificity (>0.80) but variable sensitivity across languages and review types.

For data extraction, Gartlehner et al. [2024] showed in *Research Synthesis Methods* that Claude 2 achieved 96.3% accuracy compared to human reviewers, with test-retest reliability of 95–97%. Jensen et al. [2025] found that ChatGPT-4o achieves 92.4% extraction accuracy and 94.1% inter-session reproducibility in *PLOS ONE*, explicitly noting residual stochasticity across sessions.

More comprehensive pipelines have also been proposed. The TrialMind system [Wang et al., 2025], published in *npj Digital Medicine*, demonstrated that human-AI collaboration improved recall by 71.4% and reduced screening time by 44.2% across 100 systematic reviews. Li et al. [2025a] described an LLM pipeline for health technology assessment in *JAMIA*, while Matsui et al. [2024] showed that a three-layer GPT-3.5/GPT-4 strategy achieved human-comparable sensitivity in *JMIR*.

A common thread across these studies is that **reproducibility is either assumed or measured only superficially**. Test-retest reliability, when reported, is based on 2–3 repetitions and does not examine how variation propagates through multi-stage pipelines.

2.2 The Non-Determinism Problem

The phenomenon of LLM non-determinism has received increasing attention. Atil et al. [2024] provided the most comprehensive analysis to date, documenting accuracy swings of up to 15% across repeated runs and introducing the TARr@N metric to quantify run-to-run stability. Klishevich et al. [2025] confirmed that temperature = 0 fails to ensure determinism across GPT-4o, Claude 3.5, and LLaMA 3.2 on code review tasks.

At the token level, Fu et al. [2026] showed that apparent text-level agreement masks substantial distributional instability in token probabilities—a finding that suggests surface-level reproducibility metrics may underestimate true variation. The analysis by Angermeir et al. [2025] examined how non-determinism undermines reproducibility in software engineering research, proposing documentation standards to mitigate this.

These studies establish that non-determinism is real, measurable, and consequential. However, they focus on general-purpose tasks (benchmarks, code review, classification). Our study is the first to measure non-determinism in the specific context of evidence synthesis, where the structured and domain-specific nature of outputs creates unique challenges.

2.3 Reproducibility in AI Research

The broader AI reproducibility crisis has been documented extensively. Hutson [2018] named the problem in *Science*, identifying unpublished code and sensitivity to training conditions as primary barriers. Haibe-Kains et al. [2020] called for structural reform in *Nature*, advocating for mandatory code and model sharing. Kapoor & Narayanan [2023] documented data leakage affecting 294 studies across 17 fields in *Patterns*, while Han [2025] analyzed reproducibility failures in biomedical AI through the lens of model non-determinism.

Recent frameworks have attempted to systematize reproducibility requirements. Desai et al. [2025] proposed a taxonomy of reproducibility types in *AI Magazine*, arguing that LLMs require "conceptual replicability" given their proprietary training data. Semmelrock et al. [2025] provided a systematic overview of barriers in *AI Magazine*, and Yu, B. [2024] argued in the *Harvard Data Science Review* that computational reproducibility alone is insufficient for trustworthy AI.

2.4 Provenance and Transparency

Provenance tracking—recording the lineage of data, computations, and decisions—has been identified as a key enabler of AI reproducibility. Longpre et al. [2024a] audited 1,800+ datasets in *Nature Machine Intelligence*, finding license omission rates above 70%. Liang et al. [2024] analyzed 32,111 model cards on Hugging Face, finding that documentation of limitations and environmental impact remains sparse. The Foundation Model Transparency Index [Bommasani et al., 2024] scored 14 major developers across 100 indicators, finding average transparency of only 58/100.

Technical solutions for provenance include the Workflow Run RO-Crate [Leo et al., 2024] and the W3C PROV ontology [Kale et al., 2023]. Longpre et al. [2024b] argued at ICML 2024 that current practices for data authenticity and provenance are "fundamentally broken," calling for comprehensive reform.

Our provenance approach builds on the lightweight protocol proposed in Rover [2026], using SHA-256 hashing of inputs, outputs, and configurations to create an auditable trail without requiring infrastructure beyond standard file storage.

2.5 Positioning: What Existing Studies Do Not Address

Table 1 synthesizes the landscape of prior work and highlights the methodological gaps that motivate our study. Three critical gaps emerge. **First**, studies that evaluate LLMs for evidence synthesis [Oami et al., 2024, Gartlehner et al., 2024, Jensen et al., 2025, Matsui et al., 2024] measure accuracy against human judgments but do not measure self-consistency across repeated runs—or do so with only 2–3 repetitions and per-field metrics that can mask whole-output variation. **Second**, studies that measure LLM non-determinism [Atil et al., 2024, Klishevich et al., 2025, Fu et al., 2026] do so on general-purpose benchmarks (classification, code review), not on the structured, domain-specific tasks characteristic of evidence synthesis. **Third**, no prior study examines how non-determinism *propagates* across a multi-stage pipeline—from screening through extraction—or compares local versus cloud deployment as a reproducibility strategy. Our study is the first to address all three gaps simultaneously.

Table 1: Comparison of our study with prior work. "Runs" indicates the number of repeated executions under identical conditions. "Full pipeline" indicates whether both screening and extraction were evaluated in the same study. Shading highlights our study's distinguishing features.

Study	Domain	Task	Runs	Metric	Local	Cloud	Pipeline
Oami (2024)	Health	Screening	1	Accuracy	—	✓	—
Khraisha (2024)	Health	Scr. + Extr.	1	Accuracy	—	✓	—
Gartlehner (2024)	Health	Extraction	2–3	Field acc.	—	✓	—
Jensen (2025)	Health	Extraction	2	Field acc.	—	✓	—
Matsui (2024)	Health	Screening	1	Accuracy	—	✓	—
Atil (2024)	General	NLP bench.	10+	Accuracy var.	—	✓	—
Klishevich (2025)	Software	Code review	5+	Determinism	✓	✓	—
Fu (2026)	General	Token dist.	—	Distributional	—	✓	—
This study	**Health**	**Scr. + Extr.**	**10**	**EMR + CI**	✓	✓	✓

3 Methods

3.1 Study Design

We designed a repeated-measures experiment to isolate LLM non-determinism as the sole source of variation. The same corpus, prompts, model configurations, and random seeds were used across all runs. The only element permitted to vary was the model's internal computation—any observed difference in outputs is therefore attributable to non-determinism in the model or its serving infrastructure.

Figure 1 provides an overview of the experimental pipeline.

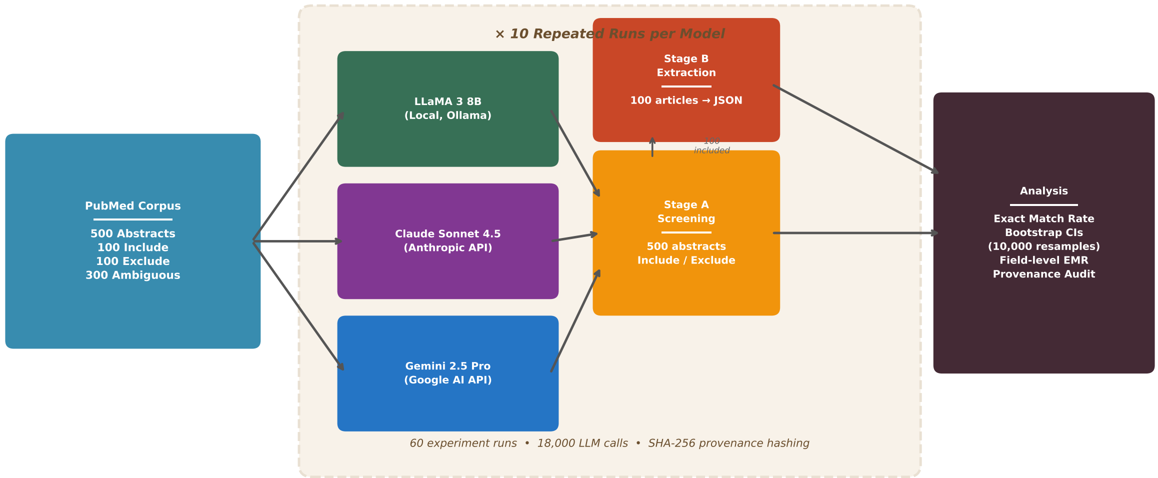

Figure 1: Experimental pipeline. A 500-abstract PubMed corpus is processed by three LLMs (one local, two cloud-based APIs), each run 10 times under identical configurations. Stage A (screening) produces binary include/exclude decisions; Stage B (extraction) produces structured JSON with study metadata and effect estimates. All outputs are hashed (SHA-256) for provenance tracking. Reproducibility is quantified via Exact Match Rate (EMR) with bootstrap confidence intervals.

3.2 Corpus Construction

We constructed a 500-abstract corpus from PubMed covering the association between $PM_{2.5}$ and respiratory health outcomes (Table 2). The domain of fine particulate matter and respiratory health was chosen because it represents one of the most extensively studied environmental health topics, with Yu, W. et al. [2024] estimating approximately 1 million premature deaths annually from short-term $PM_{2.5}$ exposure in *The Lancet Planetary Health*. This literature is characterized by standardized reporting of relative risks with confidence intervals, well-defined exposure metrics, and time-series study designs—making it an ideal testbed for evaluating LLM extraction reproducibility [Iyer et al., 2025, Yu, M. et al., 2025].

Table 2: Corpus composition for the $PM_{2.5}$ and respiratory health abstract screening task.

Category	Count	Purpose
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	**500**	

Inclusion criteria required: (1) original epidemiological study, (2) $PM_{2.5}$ as exposure,

(3) respiratory outcome (hospitalization, ED visits, or mortality), (4) time-series or case-crossover design, (5) quantitative risk estimates, and (6) peer-reviewed publication. Articles were sourced from PubMed using the E-utilities API with a broad query yielding 573 candidate records, supplemented by 475 exclusion candidates and a design-filtered subset of 222 records. The final corpus spans 148 journals and publication years 1994–2026, with a mean abstract length of 1,873 characters.

A gold standard was established using dual-independent labeling with discordance resolution for the 200 clear-category abstracts. The 300 ambiguous abstracts were classified using keyword matching and study design indicators; since our primary metric (EMR) measures self-consistency rather than accuracy, gold standard quality affects only the secondary accuracy analysis.

3.3 Models

We selected three models representing local and cloud-based deployment (Table 3).

Table 3: Model configurations. Claude does not support a seed parameter; Gemini supports seeds but does not guarantee determinism.

Model	**Provider**	**Type**	**Temperature**	**Seed**
LLaMA 3 8B	Ollama (local)	Local	0	42
Claude Sonnet 4.5	Anthropic API	Cloud	0	—
Gemini 2.5 Pro	Google AI API	Cloud	0	42

LLaMA 3 [Dubey et al., 2024] was served locally via Ollama v0.15.5 on Apple M4 hardware (24 GB RAM), ensuring single-device deterministic inference. Claude Sonnet 4.5 [Anthropic, 2024] was accessed through the Anthropic API; Claude does not support a seed parameter. Gemini 2.5 Pro [Comanici et al., 2025] was accessed through the Google AI API with seed parameter set to 42. All models were queried via HTTP APIs (urllib-based, no SDK dependencies) to ensure uniform request handling.

3.4 Pipeline Stages

Each model was run 10 times through two stages:

Stage A: Abstract Screening. Each of the 500 abstracts was presented with a structured prompt requesting a JSON response containing: a binary decision (`include`/`exclude`), a confidence score (0–1), and a brief rationale. Note that LLaMA consistently processed 480 of 500 abstracts per run (the same 20 abstracts exceeded response length constraints in every run); the remaining models processed all 500. Total: 3 models $\times$ 10 runs $\times$ 500 abstracts $=$ 15,000 screening calls.

Stage B: Data Extraction. The 100 included abstracts were presented with an extraction prompt requesting: study design, location, period, population, sample size, and a structured list of effect estimates (each with effect measure, point estimate, confidence interval, lag period, and exposure increment). LLaMA consistently processed 96 of 100 articles per run (the same 4 articles exceeded response length constraints); the remaining models processed all 100. Total: 3 models $\times$ 10 runs $\times$ 100 articles $=$ 3,000 extraction calls.

In total, the experiment comprised **18,000 LLM calls** across **60 experiment runs**.

3.5 Reproducibility Metrics

3.5.1 Exact Match Rate (EMR)

Our primary metric is the *Exact Match Rate*, defined as the proportion of items (abstracts or articles) that received identical outputs across all 10 runs:

$$\text{EMR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\,|\{o_{i,1}, o_{i,2}, \ldots, o_{i,R}\}| = 1\,] \tag{1}$$

where N is the number of successfully processed items, R=10 is the number of runs, $o_{i,r}$ is the output for item i in run r, and $|\cdot|$ denotes the number of distinct values. For screening, $o_{i,r}$ is the binary decision; for extraction, $o_{i,r}$ is the SHA-256 hash of the full JSON output (used as a compact identity check: any character-level difference produces a different hash).

An EMR of 1.0 indicates perfect reproducibility; an EMR of 0.0 indicates that every item received at least one different output across runs. We note that extraction EMR is a strict metric—any variation in any field (including formatting differences such as "Beijing, China" vs. "Beijing") produces a non-match. The field-level analysis in Section 4.4 complements this by isolating which fields drive the variation.

3.5.2 Bootstrap Confidence Intervals

We computed 95% confidence intervals for EMR using the percentile bootstrap method with 10,000 resamples. For each bootstrap sample, we drew N items with replacement and computed the EMR, then reported the 2.5th and 97.5th percentiles of the bootstrap distribution.

3.5.3 Secondary Metrics

- **Flip rate**: $1 - \text{EMR}$ (proportion of items with any variation across runs)

- **Mean agreement**: Average proportion of runs agreeing with the majority decision per item
- **Accuracy vs. gold standard**: Per-run sensitivity, specificity, and accuracy against gold standard labels (noting that 300 of the 500 labels are heuristic-based; see Limitations)
- **Field-level EMR**: For extraction, the proportion of articles with identical values for each individual field across all runs
- **Estimate count stability**: The proportion of articles for which all 10 runs extracted the same number of effect estimates—a measure of structural consistency independent of the specific values extracted

3.6 Provenance Protocol

Following Rover [2026], we implemented a lightweight provenance protocol that creates a cryptographic fingerprint of every input-output pair, enabling automated detection of any change across runs:

- **Call-level hashing**: SHA-256 hash of (prompt template || input text || model ID || temperature || seed) for each API call
- **Output hashing**: SHA-256 hash of the raw response text
- **Run cards**: Per-run metadata including timestamps, token counts, success/failure rates, and aggregate output hashes

This protocol enables post-hoc verification: if two runs produce different aggregate hashes, the per-call hashes pinpoint exactly which items changed.

3.7 Software and Reproducibility

All code is available at `https://github.com/Roverlucas/llm-evidence-synthesis-reproducibili`
The experiment was conducted in Python 3.14 on macOS (Apple M4, 24 GB RAM). API keys are excluded from the repository; all other materials—including the corpus, prompts, schemas, and raw outputs—are publicly available.

4 Results

4.1 Overview

All 60 experiment runs completed successfully: 30 screening runs (10 per model × 500 abstracts each) and 30 extraction runs (10 per model × 100 articles each), totaling 18,000 LLM calls. LLaMA consistently processed 480/500 screening abstracts and 96/100 extraction articles across all 10 runs—the same items failed every time due to response length constraints, confirming that the failures were deterministic. EMR for LLaMA was therefore computed over the 480 (screening) and 96 (extraction) successfully processed items. Claude achieved 500/500 and 100/100 in all runs. Gemini achieved 483–500/500 screening and 98–100/100 extraction.

4.2 Screening Reproducibility

Table 4 presents the screening reproducibility results.

Table 4: Screening reproducibility across 10 runs. LLaMA EMR is computed over the 480 consistently successful abstracts.

Model	EMR	95% CI	Flip Rate	Accuracy	Sensitivity	Specificity
LLaMA 3 8B	1.000	[1.000, 1.000]	0.0%	0.927	0.928	0.926
Claude Sonnet 4.5	0.974	[0.960, 0.986]	2.6%	0.931	0.862	1.000
Gemini 2.5 Pro	0.936	[0.914, 0.956]	6.4%	0.964	0.929	1.000

LLaMA produced identical screening decisions across all 10 runs for every successfully processed abstract—perfect reproducibility within its effective sample. Claude showed 2.6% flip rate (13 abstracts changed their decision at least once across runs), while Gemini showed 6.4% flip rate (32 abstracts).

Notably, accuracy and reproducibility behaved as independent properties: Gemini achieved the highest accuracy (0.964) but the lowest reproducibility (EMR=0.936), while LLaMA had the lowest accuracy (0.927) but perfect reproducibility. This illustrates that a model can be accurate on average yet inconsistent across individual runs.

4.3 Extraction Reproducibility

The extraction results reveal a dramatically larger reproducibility gap (Table 5).

Table 5: Extraction reproducibility across 10 runs. LLaMA EMR is computed over the 96 consistently successful articles.

Model	EMR	95% CI	Estimate Stability
LLaMA 3 8B	1.000	[1.000, 1.000]	1.000
Claude Sonnet 4.5	0.050	[0.010, 0.090]	0.880
Gemini 2.5 Pro	0.200	[0.120, 0.280]	0.735

While screening EMR remained above 0.93 for all models, extraction EMR dropped to **0.050 for Claude** (only 5 of 100 articles produced identical extraction outputs across all 10 runs) and **0.200 for Gemini** (20 of 100). LLaMA again achieved perfect reproducibility across all successfully processed articles.

We emphasize that extraction EMR is computed over the full JSON output hash, so any character-level variation counts as a mismatch. To assess practical significance, we examine field-level variation below.

4.4 Field-Level Analysis

Not all extraction fields are equally affected (Table 6).

Table 6: Field-level EMR for extraction outputs.

Field	LLaMA	Claude	Gemini
Study design	1.000	0.990	1.000
Study period	1.000	0.990	1.000
Sample size	1.000	0.970	1.000
Population	1.000	0.960	0.948
Study location	1.000	0.780	0.896
Estimate count stability	**1.000**	**0.880**	**0.735**

A clear pattern emerges: **categorical fields** (study design, study period) are highly reproducible even for API models (EMR$\geq$0.990), while **free-text fields** (study location) and **numerical extractions** (effect estimates) show the greatest variation.

The estimate count stability metric reveals that 26.5% of Gemini's articles and 12.0% of Claude's articles received a different number of effect estimates across runs—a structural difference that would directly affect any downstream meta-analysis.

4.5 Semantic Equivalence Analysis

The hash-based EMR reported in Table 5 (0.050 for Claude, 0.200 for Gemini) treats any character-level difference in the *full JSON output*—including all numeric effect estimates, confidence intervals, and variable-length estimate arrays—as a mismatch. To disentangle variation in the five metadata fields from variation in numeric estimates, and to assess whether metadata variation is merely cosmetic (formatting, abbreviation) or genuinely semantic, we applied three progressively relaxed equivalence criteria to the metadata fields only (Table 7).

1. **Exact match**: character-identical outputs (equivalent to SHA-256 hash comparison).

2. **Normalized match**: after lowercasing, whitespace normalization, punctuation stripping, and abbreviation expansion (e.g., "U.S." $\rightarrow$ "United States").

3. **Fuzzy match**: Levenshtein similarity ratio ≥ 0.90 between all normalized pairwise comparisons across runs.

4. **BERTScore**: Embedding-based semantic similarity using `roberta-large`, computing the F1 score across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model).

Table 7: Semantic equivalence under progressively relaxed criteria ($n = 100$ articles, 10 runs). Columns 2–4 report whole-output EMR over the five metadata fields plus estimate count stability. BERTScore F1 (`roberta-large`) reports the mean $\pm$ SD across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model). Hash-based EMR (from Table 5) additionally requires identical numeric estimates. The gap between high BERTScore ($\geq$0.997) and low hash-based EMR ($\leq$0.200) demonstrates that even near-identical metadata co-occurs with divergent numeric extractions.

Model	Exact	Norm.	Fuzzy	BERTScore F1	Hash*
LLaMA 3 8B	1.000	1.000	1.000	1.000 $\pm$ 0.000	1.000
Claude Sonnet 4.5	0.640	0.680	0.690	0.997 $\pm$ 0.009	0.050
Gemini 2.5 Pro	0.620	0.620	0.630	0.997 $\pm$ 0.022	0.200

*From Table 5; includes all fields and numeric estimates.

Three findings emerge from this multi-level analysis.

First, the gap between metadata-field EMR (0.640 for Claude) and hash-based EMR (0.050) quantifies how much variation resides in the numeric estimate arrays versus the text metadata. For Claude, 64% of articles had identical metadata across all runs, but only 5% had identical *complete* outputs—meaning that the variable-length estimate arrays (effect sizes, confidence intervals) are the primary source of non-reproducibility.

Second, within the metadata fields, normalization and fuzzy matching improved EMR by only 1–5 percentage points (Claude: 0.640 $\rightarrow$ 0.690; Gemini: 0.620 $\rightarrow$ 0.630), confirming that the variation is predominantly semantic, not cosmetic. Study location—the most variable field—improved modestly under fuzzy matching (Claude: 0.780 $\rightarrow$ 0.830; Gemini: 0.860 $\rightarrow$ 0.870), while categorical fields remained unchanged at $\geq$0.960.

Third, BERTScore embedding similarity computed across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model) was near-perfect for API models: Claude averaged F_1=0.997 $\pm$ 0.009 and Gemini F_1=0.997 $\pm$ 0.022, with 99.7% and 98.5% of pairs exceeding $F_1 \geq 0.95$, respectively. This creates an apparent paradox: the models produce *semantically near-identical* metadata text, yet 95% of Claude's and 80% of Gemini's complete extraction outputs are non-identical. The resolution lies in the numeric estimates—the effect sizes, confidence intervals, and variable-length estimate arrays that drive the hash-based EMR to 0.050 and 0.200. Even when a model describes a study in nearly the same words, it may extract a different number of effect estimates or parse "RR = 1.05 (95% CI: 1.01–1.09)" with subtly different numeric precision. Notably, Gemini's higher standard deviation (0.022 vs. 0.009 for Claude) and lower minimum BERTScore (F_1=0.754) indicate that some articles receive genuinely divergent interpretations across runs—not merely numeric rounding differences.

4.6 Meta-Analytic Propagation

To illustrate how extraction non-determinism propagates into downstream analysis, we performed a fixed-effect inverse-variance meta-analysis on the extracted risk ratios (RR/OR/HR) from each run independently (Table 8).

Table 8: Meta-analytic inputs across 10 runs. "k" is the number of usable effect estimates per run; "k_{sig}" is the number of those estimates whose confidence interval excludes the null. Δk and Δk_{sig} report the range across runs. The "Varying" column reports the proportion of articles for which the number of extracted estimates changed across runs.

Model	k **range**	Δk	k_{sig} **range**	Δk_{sig}	**Varying**
LLaMA 3 8B	246–246	0	188–188	0	0/100 (0%)
Claude Sonnet 4.5	186–209	23	152–175	23	21/100 (21%)
Gemini 2.5 Pro	154–171	17	133–150	17	20/100 (20%)

Two findings emerge. **First**, one in five articles produced a different number of effect estimates across runs for both API models. Claude's per-run estimate count varied from 186 to 209 (a 12.4% swing), while Gemini's varied from 154 to 171 (11.0%). LLaMA produced identical sets of 246 estimates in every run.

Second, the variation in estimate counts directly mirrors the variation in significant estimates—every estimate gained or lost between runs is one that crosses or retreats from the significance threshold. For Claude, the number of significant estimates ranged from 152 to 175 (a swing of 23), meaning that **up to 23 study-level effect estimates—each representing a distinct epidemiological finding—appear or disappear from the evidence base depending solely on which run of the LLM was used**.

While the overall pooled effect in our corpus remained stable across runs (because the aggregate signal is strong and k is large), the instability of the underlying evidence composition is alarming. In domains with smaller literatures or marginal effects, this level of input instability could plausibly shift pooled conclusions across the significance threshold—a hypothesis that merits direct testing in future work.

4.7 Summary Comparison

Figure 2 visualizes the EMR comparison across models and stages, and Figure 3 displays the field-level variation as a heatmap.

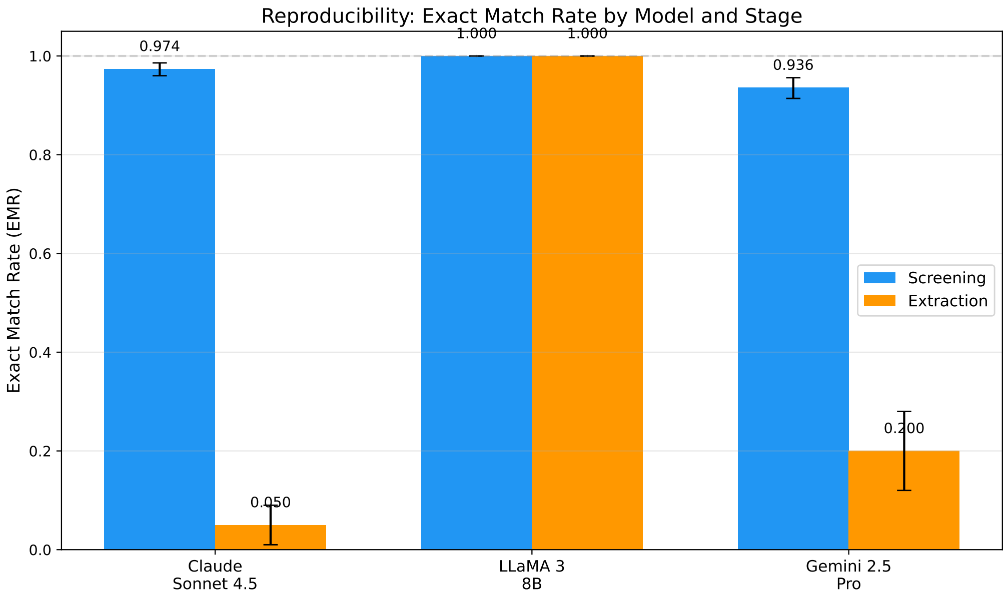

Figure 2: Exact Match Rate (EMR) by model and pipeline stage. Error bars represent 95% bootstrap confidence intervals. The dashed line indicates perfect reproducibility (EMR=1.0).

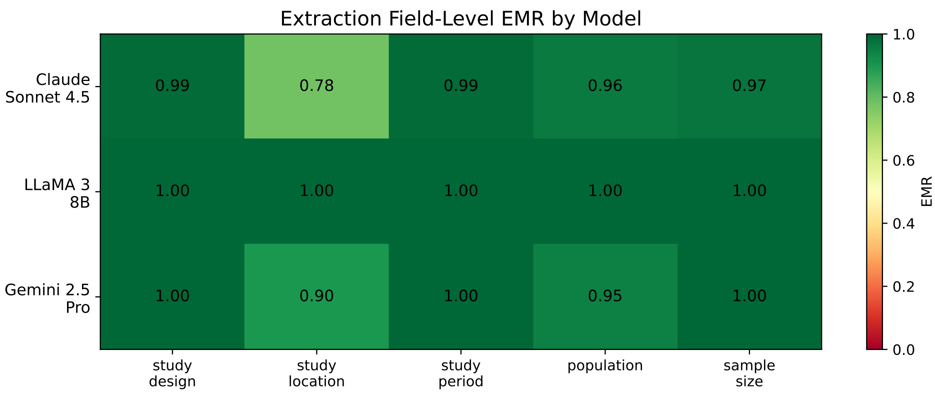

Figure 3: Field-level EMR heatmap for extraction outputs. Darker cells indicate lower reproducibility. Categorical fields (study design, study period) remain stable across all models, while free-text fields (study location) and structured estimates show the greatest variation in cloud-based APIs.

The results reveal two key patterns: (1) a sharp reproducibility gap between local and cloud-based deployment, and (2) a dramatic amplification of non-determinism when moving from simple binary screening decisions to complex structured extraction outputs.

5 Discussion

5.1 Principal Findings

Our study provides the first systematic quantification of LLM non-determinism across a complete evidence synthesis pipeline. Three key findings emerge.

Finding 1: Local deployment achieves deterministic inference; cloud APIs do not. LLaMA 3 8B, deployed locally via Ollama on a single device, produced identical outputs across all 10 runs in both screening and extraction. Both cloud-based APIs (Claude, Gemini) exhibited measurable non-determinism despite requesting temperature $= 0$ and fixed seeds. This gap has a precise technical explanation rooted in floating-point arithmetic. Cloud APIs serve requests across multi-GPU clusters where matrix multiplications are partitioned and reduced in non-deterministic order. Because floating-point addition is not associative—$(a + b) + c \neq a + (b + c)$ in general due to rounding—different partitioning strategies produce subtly different intermediate sums, which propagate through the autoregressive generation process and can eventually cause different token selections [Karelin et al., 2024]. Local deployment on a single device with a single execution thread avoids this source of variation entirely: the same computation graph executes in the same order every time, producing bit-identical results. We note, however, that this determinism guarantee is configuration-specific: the same model on different hardware (e.g., different GPU architectures with different floating-point unit implementations), different driver versions, different Ollama versions, or with parallelized inference (e.g., tensor parallelism across multiple GPUs) would likely produce different baseline outputs—deterministic within each configuration, but not identical across configurations.

Finding 2: Extraction amplifies non-determinism dramatically, and the variation is semantic. Claude's EMR dropped from 0.974 (screening) to 0.050 (extraction); Gemini's from 0.936 to 0.200. Screening requires a binary decision (a single token effectively determines the output), while extraction generates complex structured JSON with multiple fields. Our four-tier semantic equivalence analysis (Section 4.5) confirms that this gap is not an artifact of strict string matching. BERTScore embedding similarity across all 4,500 pairwise comparisons per model averaged F_1=0.997—near-perfect semantic agreement on metadata text—yet hash-based EMR remained at 0.050 (Claude) and 0.200 (Gemini). This paradox reveals that the variation concentrates in the numeric extraction layer: the models describe studies in nearly the same words but extract different numbers of effect estimates and parse numeric values with different precision. Even the residual metadata variation is genuinely semantic (fuzzy matching recovers only 1–5 percentage points), not cosmetic.

Finding 3: Accuracy and reproducibility are independent properties. Gemini achieved the highest screening accuracy (0.964) but the lowest screening reproducibility (0.936). This independence means that **accuracy evaluations from a single run cannot detect reproducibility problems**—a model that performs well in one evaluation may produce materially different results in a replication attempt.

5.2 Implications for Evidence Synthesis Practice

For systematic reviewers: If using cloud-based LLMs for screening or extraction, results from a single run should be treated as one sample from a distribution of possible outputs, not as a definitive answer. Running multiple repetitions and reporting variation metrics should become standard practice, analogous to how human inter-rater reliability is routinely measured.

For journal editors and reviewers: Papers reporting LLM-assisted evidence synthesis should be expected to disclose: (a) the exact model version, (b) all inference parameters, (c) whether results were verified across multiple runs, and (d) provenance information enabling independent replication.

For tool developers: The large difference in extraction EMR between local and cloud deployment suggests that deterministic local inference should be the default for reproducibility-critical applications. Where cloud APIs must be used (e.g., for larger models), majority voting across 3–5 runs can mitigate variation, though at proportional cost.

5.3 Relationship to Prior Work

Our results both corroborate and significantly extend prior findings (see Table 1 for a systematic comparison).

Our screening flip rates (2.6% for Claude, 6.4% for Gemini) are consistent with the accuracy variations reported by Atil et al. [2024] on general benchmarks, confirming that the phenomenon documented in controlled settings translates directly to real-world evidence synthesis tasks. However, we reveal for the first time that **structured extraction amplifies non-determinism by an order of magnitude**—a finding invisible to benchmark-level analyses. Claude's EMR dropped from 0.974 (screening) to 0.050 (extraction)—a 19.5× amplification that has no precedent in the literature.

The comparison with Jensen et al. [2025] is particularly instructive. Their reported 94.1% inter-session reproducibility for ChatGPT-4o extraction appears to align with our Claude screening EMR (97.4%), creating the impression that API-based extraction is reasonably stable. However, their metric measures per-field agreement across 2 sessions, while our EMR measures whole-output identity across 10 runs. When we apply our stricter metric, extraction reproducibility drops to 5.0%—**nearly twenty times lower than what per-field metrics suggest**. This discrepancy is not an artifact; it reflects a genuine measurement gap. Per-field metrics miss the combinatorial explosion of variation: when 5 fields each have 97% individual reproducibility, the probability that *all* fields match simultaneously is only $0.97^5 = 0.859$—and this is before accounting for the variable-length effect estimate lists that drive most of our observed non-matches.

Similarly, Gartlehner et al. [2024]'s 95–97% test-retest reliability with Claude 2 was measured over 2–3 repetitions. Our 10-run design with output hashing reveals instability that shorter test-retest windows cannot detect, because some items change in only 1–2 of 10 runs—a frequency that 2-run comparisons would miss entirely.

The meta-analytic propagation experiment (Section 4.6) provides the most direct evidence of downstream impact. For Claude, the number of usable effect estimates varied from 186 to 209 across runs—a 12.4% swing in the evidence base feeding the pooled result. For Gemini, the swing was 11.0% (154–171 estimates). One in five articles produced a different number of effect estimates depending on the run, meaning that **the composition of a meta-analysis is not fixed when an LLM extracts the data**. While the overall pooled conclusion was stable in our corpus (because the aggregate effect was strong and k was large), this level of compositional instability raises serious concerns for systematic reviews in domains with smaller literatures or marginal effects, where even a few added or dropped estimates could shift a confidence interval across the null.

5.4 Limitations and Future Directions

We frame each limitation as a foundation for future research.

Model coverage. We tested three models representing two deployment paradigms (local vs. cloud). A natural extension is to benchmark additional models (GPT-4o, Mistral, Command-R, open-weight models of varying sizes) to map the reproducibility landscape more comprehensively and identify whether non-determinism scales with model size or architecture.

Run count and rare events. Ten repetitions provide adequate statistical power for EMR estimation with bootstrap CIs, but may not capture rare variation patterns that appear in <10% of runs. Our prior work [Rover, 2026] used up to 100 repetitions; scaling to 50–100 runs with cost-efficient models would enable detection of tail-event instability.

Domain generalization. Our corpus focuses on $PM_{2.5}$ and respiratory health, a domain with standardized reporting conventions. Cross-domain studies—particularly in behavioral health, pharmacology, or nutritional epidemiology where reporting is more heterogeneous—would test whether the amplification patterns we observed are domain-specific or universal.

Semantic equivalence thresholds. Our four-tier semantic analysis (Section 4.5), including BERTScore embedding similarity, demonstrated that metadata variation is predominantly semantic rather than cosmetic. Future work should establish *clinically meaningful* equivalence thresholds—e.g., what BERTScore difference in an extraction output corresponds to a materially different meta-analytic conclusion?—and validate these thresholds across multiple health domains.

Meta-analytic significance flipping. Our meta-analytic propagation experiment (Section 4.6) showed that up to 23 study-level estimates appear or disappear between runs. While the pooled conclusion was stable in our large-k corpus, directly testing whether significance flips occur in smaller literatures (e.g., $k<20$) with marginal effects is an important next step for understanding real-world impact.

Screening EMR scope. Our screening EMR considers only the binary include/exclude decision, yet the output also contains confidence scores, rationales, and sub-classifications that may vary even when the binary decision is stable. Full-output screening EMR would provide a more complete picture of screening-stage non-determinism.

API versioning and temporal stability. Cloud API models can be silently updated during multi-day experiments. While we pinned Claude to a dated version (`claude-sonnet-4-5-20250929`), Gemini's version identifier does not include a date suffix. Longitudinal studies tracking EMR over weeks or months would distinguish inference-level non-determinism from model update effects—a distinction critical for establishing stable benchmarks.

5.5 Recommendations

Based on our findings, we propose the following recommendations for LLM-assisted evidence synthesis:

1. **Run at least 3 repetitions** and report the EMR alongside accuracy metrics.
2. **Prefer local deployment** for reproducibility-critical tasks when model capability permits.
3. **Implement provenance hashing** to enable post-hoc verification and auditing.
4. **Report full model specifications**: model ID, version, temperature, seed, and any API-specific parameters.

5. **Use majority voting** when local deployment is not feasible, accepting the cost trade-off for improved stability.

6. **Distinguish accuracy from reproducibility** in evaluations—a model can be accurate on average but unreliable for any single run.

6 Conclusion

We have shown that LLM non-determinism is not a theoretical concern but a measurable, quantifiable phenomenon that directly threatens the reproducibility of evidence synthesis. Through 18,000 API calls across 60 experiment runs—the most comprehensive reproducibility evaluation of LLMs for evidence synthesis to date—we demonstrate four findings with immediate practical implications.

First, local deployment achieves perfect reproducibility, proving that deterministic LLM inference is technically achievable. Second, cloud-based APIs produce variable outputs even under "deterministic" settings, with extraction tasks amplifying this instability by an order of magnitude: while prior work reports 94–97% reproducibility using per-field metrics and 2–3 runs, our whole-output analysis across 10 runs reveals that only 5–20% of extraction outputs are truly identical. Third, semantic equivalence analysis—including BERTScore embedding similarity (F_1=0.997)—reveals a paradox: models produce near-identical metadata text across runs, yet extract different numeric values and different numbers of effect estimates, driving hash-based EMR to 5–20%. Fourth, meta-analytic propagation analysis reveals that one in five articles produces a different number of effect estimates across runs, causing the composition of a meta-analysis to depend on which LLM run generated its inputs.

These results carry a clear message: the current standard of reporting LLM-assisted evidence synthesis results from a single, undocumented run is scientifically insufficient. As LLMs become embedded in systematic reviews and meta-analyses, the research community must adopt reproducibility-aware practices—multiple repetitions, provenance tracking, and transparent reporting of variation—or risk building an evidence base on foundations that shift each time the analysis is re-run.

Author Contributions

LR conceived and designed the study, developed the software pipeline, conducted all experiments, performed the statistical analysis, and wrote the manuscript. YST supervised the research, provided critical feedback on the methodology and interpretation of results, and reviewed the manuscript.

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Conflict of Interest

The author declares no conflicts of interest.

Data Availability Statement

All code, data, prompts, schemas, raw outputs, and run provenance records are publicly available at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility. API keys are excluded from the repository. The 500 PubMed abstract identifiers, gold standard labels, and complete analysis scripts are included to enable full reproduction of the reported results.

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